

Name of the Student	
Reg. No	
GitHub Link	

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 3

Class Test

COURSE NAME: Deep Learning
Faculty : Dr. Arul Raj A.M

COURSE CODE: MLA04
Slot :

COURSE OUTCOME COVERED

CO 2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage : 15%

Assessment Tool Weightage: 20%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Class Test

CO2-AT3

QUESTIONS: 05

Total Marks:25

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of **Representation Learning**. Discuss how the **width** and **depth** of a neural network influence its learning capability, computational complexity and model performance.
3. Compare **ReLU**, **Leaky ReLU (LReLU)** and **ELU (Exponential Linear Unit)** activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of **unsupervised learning** in neural networks. Describe the working principle of **Restricted Boltzmann Machines (RBMs)** and **Autoencoders** and compare their objectives and applications.
5. Explain the concept of Autoencoder and its uses in Deep learning .

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand
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